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## Linear Statistical Models

B. Statistical Data Science 2nd Year      Indian Statistical Institute

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### Exercise Series 2

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**Exercise 1.** Consider a simple linear regression model

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, \dots, n,$$

with  $\mathbb{E}(\epsilon_i) = 0$ ,  $\text{Var}(\epsilon_i) = \sigma^2$  for all  $i = 1, \dots, n$ , and  $\text{Cov}(\epsilon_i, \epsilon_j) = 0$  for all  $1 \leq i \neq j \leq n$ . Recall that the (ordinary) least-squares estimators of  $\alpha$  and  $\beta$  are obtained by minimizing  $\sum_{i=1}^n (y_i - a - bx_i)^2$  w.r.t.  $a$  and  $b$ . We saw in the class that the least-squares estimators are given by

$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and} \quad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}.$$

We also saw that  $\mathbb{E}(\hat{\alpha}) = \alpha$  and  $\mathbb{E}(\hat{\beta}) = \beta$ , i.e., the estimators are unbiased.

- (a) Show that  $\text{Var}(\hat{\alpha}) = \sigma^2 \left\{ \frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right\}$  and  $\text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$ .
- (b) Suppose that  $n$  is even and the  $x_i$ 's take values in  $[a, b]$ , where  $a, b$  are known constants. If you are allowed to choose the  $x_i$ 's, how would you choose them so that  $\text{Var}(\hat{\beta})$  is minimized? What would your strategy be if  $n$  is odd?

**Note:** Choosing the covariates appropriately in an experiment is an important problem and falls under the broader class of “designs of experiments”.

**Exercise 2.** Again consider the simple linear regression setup of Exercise 1. Recall that after estimating the model parameters  $\alpha$  and  $\beta$ , our fitted regression model becomes  $\hat{y} = \hat{\alpha} + \hat{\beta}x$ . The fitted  $y$ -values are  $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ ,  $i = 1, \dots, n$ . And, the residuals are defined as  $e_i = y_i - \hat{y}_i = y_i - \hat{\alpha} - \hat{\beta}x_i$ ,  $i = 1, \dots, n$ .

- (a) Verify that  $\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i$ , so that  $\bar{\hat{y}} = \frac{1}{n} \sum_{i=1}^n \hat{y}_i = \bar{y}$ .
- (b) Show that  $\sum_{i=1}^n y_i^2 = \sum_{i=1}^n \hat{y}_i^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$ . [Equivalently,  $\sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n y_i^2 - \sum_{i=1}^n \hat{y}_i^2$ ].
- (c) Use part (a) to show that  $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$ .
- (d) Recall that the coefficient of determination is defined as  $R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$ . Use part (c) to show that  $0 \leq R^2 \leq 1$ . Identify the situations when  $R^2 = 0$  or  $R^2 = 1$ .
- (e) Show that  $R^2 = r_{xy}^2$ , where  $r_{xy}$  is the (sample) correlation coefficient between  $x$  and  $y$ .

**Note:** In the class, we saw that (a) holds if our model has an intercept term. Without the intercept, (a) may not hold. Notice that part (b) holds irrespective of whether our model has an intercept term. To get to part (c), we again require part (a) to hold, which may not be true if our model does not have an intercept.

**Exercise 3.** In the class, we discussed about the effects of having or not having an intercept in our regression model. In the following, we examine this effect theoretically. Let  $(x_1, y_1), \dots, (x_n, y_n)$  be a collection of bivariate observations. Consider two regression models of  $y$  on  $x$ , with and without the intercept term.

$$y_i = \alpha + \beta x_i + \epsilon_i \quad \text{and} \quad y_i = \gamma x_i + \varepsilon_i.$$

Let  $\hat{\beta}$  and  $\hat{\gamma}$  be the least-squares estimates of  $\beta$  and  $\gamma$  from the two models, respectively.

- (a) Show that  $\hat{\beta} = c\hat{\gamma} + d\bar{y}$ , where  $c \geq 1$  and  $d$  are constants depending on  $x_1, \dots, x_n$ . Thus, in general, the two estimators  $\hat{\beta}$  and  $\hat{\gamma}$  are different.
- (b) Determine conditions under which the two estimators are the same.
- (c) A common practice in machine learning is to *standardize* the data before fitting any model. One common way of standardization is to mean-center (i.e., subtract the mean from the data) and scale (i.e., divide the data by the corresponding standard deviation) the observations. Check what happens if instead of  $(x_i, y_i)$ , one works with the mean centered data. What about the case where the data is mean-centered as well as scaled?

**Exercise 4.** The results from the previous exercise can be generalized to multiple linear regression as well. Consider a (multiple) linear regression model with  $p$  covariates

$$y_i = \alpha + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n.$$

We write this model as  $\mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ , with  $\tilde{\mathbf{X}} = (\mathbf{1}, \mathbf{X})$ ,  $\boldsymbol{\theta} = (\alpha, \beta_1, \dots, \beta_p)^\top = (\alpha, \boldsymbol{\beta}^\top)^\top$  and  $\mathbf{X}$  is the  $n \times p$  data matrix of the covariates  $x_1, \dots, x_p$ . We know that the least-squares estimator (LSE) of  $\boldsymbol{\theta}$  is given by (assuming  $\tilde{\mathbf{X}}$  to be of full column rank  $p+1$ )  $\hat{\boldsymbol{\theta}}_{\text{LSE}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$ . From this, we get an estimator of  $\boldsymbol{\beta}$ , say  $\hat{\boldsymbol{\beta}}_1$ . Instead, one can also get an estimator of  $\boldsymbol{\beta}$  by fitting a model with no intercept

$$y_i = \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n,$$

which leads to  $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ . In the following, we will study the relationship between  $\hat{\boldsymbol{\beta}}_1$  and  $\hat{\boldsymbol{\beta}}_2$ . In order to do that, first observe that

$$\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = \begin{pmatrix} n & \mathbf{1}^\top \mathbf{X} \\ \mathbf{X}^\top \mathbf{1} & \mathbf{X}^\top \mathbf{X} \end{pmatrix} = \begin{pmatrix} n & n\bar{\mathbf{X}}^\top \\ n\bar{\mathbf{X}} & \mathbf{X}^\top \mathbf{X} \end{pmatrix}, \quad \text{where } \bar{\mathbf{X}}^\top = (\bar{X}_1, \dots, \bar{X}_p)^\top.$$

Matrices of the above form are called *partitioned matrices*.

- (a) Let  $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$  be a partitioned matrix. Recall that  $\mathbf{A}$  is invertible if and only if  $\mathbf{A}_{11}$  and  $\mathbf{B} := \mathbf{A}_{22} - \mathbf{A}_{21}^\top \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$  are invertible. Verify that in this case,

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{B}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{B}^{-1} \\ -\mathbf{B}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{B}^{-1} \end{pmatrix}.$$

In the special case when  $\mathbf{A}_{12}$  and  $\mathbf{A}_{21}$  are vectors, which is the case in our problem, we get

$$\begin{pmatrix} a_{11} & \mathbf{a}_{12}^\top \\ \mathbf{a}_{21} & \mathbf{A}_{22} \end{pmatrix}^{-1} = \frac{1}{a_{11}^2} \begin{pmatrix} a_{11} + \mathbf{a}_{12}^\top \mathbf{B}^{-1} \mathbf{a}_{21} & -a_{11} \mathbf{a}_{12}^\top \mathbf{B}^{-1} \\ -a_{11} \mathbf{B}^{-1} \mathbf{a}_{21} & a_{11} \mathbf{B}^{-1} \end{pmatrix}, \quad \text{where } \mathbf{B} = \mathbf{A}_{22} - \frac{\mathbf{a}_{21} \mathbf{a}_{12}^\top}{a_{11}}.$$

$\mathbf{B}$  is of the form  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top$ . So, we need to find the inverse of matrices of this form.

(b) Verify that  $(A + \mathbf{u}\mathbf{v}^\top)^{-1} = A^{-1} - \frac{A^{-1}\mathbf{u}\mathbf{v}^\top A^{-1}}{1 + \mathbf{v}^\top A^{-1}\mathbf{u}}$ .

**Note:** In general, for matrices  $A, B, C$ , and  $D$  of appropriate sizes, with  $A$  and  $C$  invertible, we have

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}.$$

(c) Use parts (a) and (b) to get a closed-form expression for  $\hat{\beta}_1$ .

(d) Show that  $\hat{\beta}_1 = (I + C)\hat{\beta}_2 + \bar{y}\mathbf{d}$ , for some matrix  $C$  and vector  $\mathbf{d}$  depending on  $X$ . Thus, in general,  $\hat{\beta}_1$  and  $\hat{\beta}_2$  are different.

(e) Determine conditions under which  $C = \mathbf{O}$  and  $\mathbf{d} = \mathbf{0}$ .

**Note:** The results in parts (a) and (b) can be used to find a connection between the estimators of  $\theta_2$  from the two models:

$$\mathbf{y} = X_1\theta_1 + X_2\theta_2 + \epsilon \quad \text{and} \quad \mathbf{y} = X_2\theta_2 + \eta.$$

We will see this and its implications later.

**Exercise 5** (Simulation Study 1). In this exercise, our goal is to become familiar with fitting linear models in R. We will use some simulated data for this purpose. We will start with a simple linear regression model.

- Fix a sample size  $n$ . Remember that we need to have  $n > 1$  ( $n > 2$  if the model has an intercept).
- Generate  $n$  observations for the covariate  $x$ . You may also fix them, or get them from some existing dataset. Just be mindful that the matrix  $X$  should be of full column rank. If you generate  $x$  from an absolutely continuous distribution (e.g., normal), then this is ensured with probability 1.
- Fix some constant  $\beta$  (and  $\alpha$  if you want an intercept).
- Generate the response as  $\mathbf{y} = \alpha + \beta \star \mathbf{x} + \epsilon$ , where  $\epsilon \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ . You can use `rnorm(n)` to generate  $n$  i.i.d.  $N(0, 1)$  random variables.

**Note:** Check what happens when you add a constant to a vector/matrix in R.

- Fit a linear model with  $\mathbf{y}$  as response and  $X$  as the data matrix. In R, you can use `lm` to fit a linear model. The syntax is `lm(y ~ x)`, which fits a linear model with  $x$  as the covariate plus an intercept. For fitting a model without the intercept term, you can use `lm(y ~ 0 + x)`. After fitting the model, you can use `summary(lm)` to get many other useful quantities. Check what these quantities are. Can you understand what the estimated coefficients are? What about the SSE?

**Note:** An extensive list of R functions useful for linear models is available from this link <https://cran.r-project.org/doc/contrib/Ricci-refcard-regression.pdf>.

- `summary(lm)` gives an  $F$ -statistic and the corresponding  $p$ -value. Can you understand what the test is?

Next, we fit a multiple linear regression model.

- Fix a sample size  $n$  and number of covariates  $p$ . Remember that we need to have  $n > p$  ( $n > p + 1$  for a model with an intercept).
- Generate  $n$  observations for the covariates  $(x_1, \dots, x_p)$ . Again, be mindful that the matrix  $X$  should be of full column rank. If you generate  $x_1, \dots, x_p$  from some absolutely continuous distribution, then this holds with probability 1.

- Fix some constants  $\beta_1, \dots, \beta_p$  (and  $\alpha$  if you want an intercept).
- Generate the response as  $\mathbf{y} = \alpha + \beta_1 \star \mathbf{x}_1 + \dots + \beta_p \star \mathbf{x}_p \epsilon$ , where  $\epsilon \sim \mathbf{N}_n(\mathbf{0}, \sigma^2 \mathbf{I})$ .
- To fit a multiple linear regression model in R, you can use `lm(y ~ x1 + ... + xp)`, which fits a linear model with  $x_1, \dots, x_p$  as covariates plus an intercept. As before, you can fit a model without the intercept term using `lm(y ~ 0 + x1 + ... + xp)`.
- Use `summary(lm)` to get a summary of the fitted model. Check what the quantities are. Identify the estimated coefficients and the SSE. Try to figure out the use of the  $F$ -statistic and the corresponding  $p$ -value.

**Exercise 6** (Simulation Study 2, **Graded**). In this exercise, we study the *consistency* property of the least-squares estimators in simple linear regression setup.

- Fix the sample size  $n = 25$ . Also fix two constants  $\alpha$  and  $\beta$ . Generate  $x_1, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(25, 4)$ . Treat these as fixed covariates.
- Construct  $y_1, \dots, y_n$  using  $y_i = \alpha + \beta x_i + \epsilon_i$ ,  $i = 1, \dots, n$ , where  $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, 1)$ . Fit a simple linear regression model to find estimates of  $\alpha$  and  $\beta$ .
- Repeat step (b) 1000 times to get estimates  $\hat{\alpha}_1, \dots, \hat{\alpha}_{1000}$  and  $\hat{\beta}_1, \dots, \hat{\beta}_{1000}$ . Make histograms of  $\hat{\alpha}_i$ 's and  $\hat{\beta}_i$ 's. What do you see?
- Repeat steps (a)–(c) with  $n = 50, 100, 200, 500, 1000$ . What do you observe?

**Exercise 7** (Simulation Study 3, **Graded**). In this exercise, we study the effect of the noise  $\epsilon$  on the regression estimates. We consider the simple linear regression setup. We begin by fixing the values of  $\alpha$  and  $\beta$ . We also generate 100 observations from  $\text{Normal}(5, 1)$  and designate them as  $x_1, \dots, x_{100}$ .

- Fix  $\sigma^2 = 1$ . Construct  $y$  using  $y_i = \alpha + \beta x_i + \epsilon_i$ ,  $i = 1, \dots, 100$ , where  $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, \sigma^2)$ .
- Fit a simple linear regression model to get estimates of  $\alpha$  and  $\beta$ .
- Repeat step (b) 1000 times to get 1000 estimates of  $\alpha$  and  $\beta$ . In each of these repetitions, you should regenerate the error terms, while everything else is kept fixed. Make histograms of the estimates.
- Repeat parts (a)–(c) with  $\sigma^2 = 0.001, 0.01, 0.1, 1, 10$ . What do you observe?

**Note:** For the graded simulation studies, you need to make a report with all the relevant plots and a summary of your observations/comments.